

## IRAN SHAHR, Iran, Islamic Republic of

WMO: 408790

Lat: 27.2298N

Lon: 60.7181E

Elev: 591

StdP: 94.42

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | 4.8        | 6.7 | -19.9                         | 0.7 | 14.1 | -16.8 | 0.9 | 16.1 | 9.1                   | 18.0 | 7.3 | 19.5 | 1.6                   | 0    | 0.440 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.3                   | 45.8            | 20.6 | 44.9 | 20.4 | 44.0 | 20.3 | 25.2                | 36.7 | 24.7 | 36.5 | 24.1 | 36.6 | 4.5                  | 250  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 22.1                            | 18.0 | 33.0 | 21.0 | 16.8 | 34.7 | 19.8 | 15.6 | 35.0 | 81.4          | 36.6 | 79.0 | 36.1 | 76.3 | 36.5 | 29.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 9.0               | 7.5  | 6.4 | 1.3                        | 47.7 | 2.0                | 0.6 | -0.1                                               | 48.2 | -1.3       | 48.5 | -2.4       | 48.9 | -3.9       | 49.4 |    |    |
|                   |      |     | -3.2                       | 26.2 | 1.8                | 0.6 | -4.5                                               | 26.6 | -5.5       | 27.0 | -6.5       | 27.3 | -7.8       | 27.8 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 27.4   | 15.1  | 17.8  | 22.5  | 28.1  | 33.8  | 37.4  | 38.0  | 36.3  | 33.4  | 28.3  | 21.3  | 16.5 |
|                                                                                  | DBStd       | 8.55   | 3.04  | 3.67  | 3.16  | 3.09  | 2.32  | 2.03  | 1.63  | 1.70  | 1.97  | 2.58  | 3.32  | 3.12 |
|                                                                                  | HDD10.0     | 7      | 3     | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 2    |
|                                                                                  | HDD18.3     | 240    | 104   | 49    | 4     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 12    | 71   |
|                                                                                  | CDD10.0     | 6364   | 163   | 219   | 388   | 543   | 736   | 821   | 867   | 814   | 702   | 568   | 340   | 203  |
|                                                                                  | CDD18.3     | 3554   | 5     | 33    | 133   | 293   | 478   | 571   | 609   | 556   | 452   | 310   | 102   | 14   |
|                                                                                  | CDH23.3     | 58355  | 72    | 334   | 1494  | 4128  | 8217  | 10347 | 10967 | 9833  | 7507  | 4217  | 1058  | 181  |
| CDH26.7                                                                          | 40826       | 6      | 85    | 638   | 2421  | 5812  | 7956  | 8488  | 7362  | 5205  | 2455  | 375   | 23    |      |
| Wind                                                                             | WSAvg       | 2.6    | 2.0   | 2.3   | 2.4   | 2.7   | 2.9   | 3.3   | 3.4   | 3.2   | 2.9   | 2.3   | 2.1   | 1.9  |
| Precipitation                                                                    | PrecAvg     | 116    | 22    | 23    | 23    | 5     | 2     | 6     | 8     | 9     | 1     | 3     | 3     | 14   |
|                                                                                  | PrecMax     | 243    | 166   | 89    | 76    | 22    | 6     | 83    | 76    | 45    | 12    | 30    | 21    | 98   |
|                                                                                  | PrecMin     | 23     | 0     | 0     | 3     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | PrecStd     | 60     | 31    | 26    | 21    | 6     | 2     | 16    | 16    | 10    | 3     | 7     | 6     | 25   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 27.0  | 31.4  | 36.1  | 40.1  | 45.0  | 47.1  | 46.9  | 45.8  | 43.2  | 40.0  | 33.2  | 29.1 |
|                                                                                  |             | MCWB   | 12.8  | 15.1  | 15.9  | 18.1  | 19.4  | 20.5  | 20.8  | 20.3  | 19.7  | 18.5  | 15.8  | 14.0 |
|                                                                                  | 2%          | DB     | 25.1  | 29.2  | 33.9  | 38.8  | 43.5  | 46.0  | 45.7  | 44.2  | 42.0  | 38.2  | 31.9  | 26.9 |
|                                                                                  |             | MCWB   | 13.1  | 14.5  | 15.5  | 17.8  | 19.4  | 20.6  | 20.6  | 20.1  | 19.1  | 18.2  | 15.6  | 13.2 |
|                                                                                  | 5%          | DB     | 23.8  | 27.1  | 32.1  | 37.2  | 42.2  | 45.0  | 44.9  | 43.4  | 41.0  | 36.9  | 30.6  | 25.2 |
|                                                                                  |             | MCWB   | 12.4  | 13.8  | 15.1  | 17.5  | 19.2  | 20.5  | 20.6  | 19.9  | 18.8  | 17.6  | 15.2  | 12.7 |
|                                                                                  | 10%         | DB     | 22.2  | 25.1  | 30.1  | 35.9  | 41.1  | 43.9  | 44.0  | 42.6  | 40.0  | 35.4  | 29.0  | 23.9 |
|                                                                                  |             | MCWB   | 11.8  | 12.9  | 14.6  | 17.1  | 19.0  | 20.3  | 20.6  | 19.7  | 18.6  | 17.1  | 14.7  | 12.2 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 15.8  | 17.4  | 18.7  | 21.4  | 23.7  | 25.7  | 26.0  | 25.7  | 24.4  | 21.7  | 18.8  | 16.5 |
|                                                                                  |             | MCDB   | 22.7  | 23.9  | 25.9  | 33.8  | 37.9  | 36.6  | 37.8  | 37.0  | 36.7  | 33.2  | 28.1  | 22.0 |
|                                                                                  | 2%          | WB     | 14.7  | 16.1  | 17.8  | 19.4  | 21.3  | 24.6  | 25.4  | 25.0  | 23.1  | 20.4  | 17.7  | 15.4 |
|                                                                                  |             | MCDB   | 21.6  | 24.3  | 26.2  | 34.1  | 38.5  | 36.3  | 37.0  | 36.0  | 35.6  | 33.2  | 27.8  | 22.8 |
|                                                                                  | 5%          | WB     | 13.9  | 15.2  | 17.0  | 18.6  | 20.3  | 23.8  | 24.9  | 24.3  | 22.3  | 19.3  | 16.7  | 14.4 |
|                                                                                  |             | MCDB   | 20.8  | 23.3  | 25.0  | 33.8  | 38.4  | 36.8  | 36.7  | 36.0  | 35.2  | 33.1  | 27.2  | 22.7 |
|                                                                                  | 10%         | WB     | 13.0  | 14.4  | 16.2  | 17.9  | 19.6  | 22.9  | 24.3  | 23.6  | 21.4  | 18.3  | 15.7  | 13.4 |
|                                                                                  |             | MCDB   | 20.4  | 22.8  | 25.1  | 33.0  | 38.2  | 38.5  | 37.0  | 36.2  | 35.1  | 32.8  | 26.4  | 22.0 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 12.6  | 12.6  | 13.1  | 13.5  | 13.8  | 13.1  | 12.3  | 13.2  | 13.8  | 14.0  | 13.5  | 13.6 |
|                                                                                  |             | MCDBR  | 14.4  | 14.6  | 15.5  | 14.8  | 14.6  | 13.9  | 12.9  | 13.6  | 14.2  | 14.5  | 14.3  | 14.9 |
|                                                                                  |             | MCWBR  | 7.4   | 6.5   | 6.1   | 5.6   | 5.2   | 5.4   | 5.3   | 5.7   | 5.9   | 6.3   | 6.9   | 7.8  |
|                                                                                  | 5% WB       | MCDBR  | 11.0  | 11.3  | 11.5  | 13.3  | 13.4  | 12.0  | 11.2  | 11.5  | 12.1  | 12.7  | 11.9  | 11.8 |
|                                                                                  |             | MCWBR  | 6.3   | 5.4   | 4.7   | 5.4   | 5.2   | 5.0   | 4.5   | 4.6   | 4.9   | 5.7   | 5.9   | 6.5  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.348  | 0.400 | 0.455 | 0.507 | 0.536 | 0.557 | 0.585 | 0.535 | 0.469 | 0.427 | 0.361 | 0.332 |      |
|                                                                                  | taud        | 2.365  | 2.185 | 1.997 | 1.889 | 1.835 | 1.807 | 1.775 | 1.867 | 1.988 | 2.092 | 2.316 | 2.407 |      |
|                                                                                  | Ebn at Noon | 893    | 867   | 837   | 803   | 778   | 758   | 735   | 771   | 813   | 825   | 869   | 896   |      |
|                                                                                  | Edh at Noon | 106    | 137   | 175   | 201   | 212   | 217   | 224   | 203   | 174   | 148   | 110   | 97    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 4.12   | 4.84  | 5.99  | 6.98  | 7.57  | 7.79  | 7.40  | 7.19  | 6.56  | 5.63  | 4.61  | 4.01  |      |
|                                                                                  | RadStd      | 0.27   | 0.35  | 0.34  | 0.30  | 0.20  | 0.27  | 0.25  | 0.26  | 0.15  | 0.12  | 0.14  | 0.21  |      |

## Historical Trends

|      | DBAvg                 | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-----------------------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) | Station Only          | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (1 neighbor) | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page